

Priyanshu Singh

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SUMMARY

Software Engineer specializing in AI-powered applications, backend engineering, and cloud-native development with hands-on experience building production-scale web platforms and intelligent automation systems. Skilled in FastAPI, React, Next.js, Node.js, PostgreSQL, Docker, Kubernetes, and AWS. Currently developing an AI-powered Career Agent Platform for the U.S. market using OpenAI APIs, LangGraph, and Playwright. AWS Certified Solutions Architect passionate about building scalable, secure, and high-performance software solutions.

EDUCATION

- Vellore Institute of Technology, Bhopal** Bhopal, India
B.Tech in Computer Science and Engineering (CGPA: 8.89) Sep 2023 – May 2027

SKILLS

- Languages:** C++, Python, JavaScript, TypeScript, Shell Scripting
- Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS
- Backend:** Node.js, Express.js, REST APIs, Microservices Architecture
- AWS:** EC2, EKS, ECS, ECR, Lambda, API Gateway, S3, CloudFront, Route53, IAM, VPC, RDS, DynamoDB, Secrets Manager, WAF, Shield, CloudFormation
- DevOps:** Docker, Kubernetes, GitHub Actions, Jenkins, Terraform, Ansible
- Databases:** MongoDB, PostgreSQL, MySQL
- Tools:** Git, GitHub, Linux, VS Code, Postman, AWS CLI

EXPERIENCE

- TM Cloud Solutions LLC (USA)** Remote
AI App Developer Intern June 2026 – Present
 - Developing an AI-powered Career Agent Platform for the U.S. job market that automates job discovery, resume matching, job applications, and email workflows using LLM-powered AI agents.
 - Building scalable backend services with FastAPI and PostgreSQL while integrating OpenAI APIs and LangGraph to orchestrate multi-agent AI workflows.
 - Developing modern frontend applications using Next.js, React, and TypeScript to deliver responsive and intuitive user experiences.
 - Implementing browser automation with Playwright to streamline end-to-end job application workflows and improve automation efficiency.
 - Collaborating with engineering teams on workflow design, technical research, testing, debugging, and scalable MVP architecture for a cloud-native SaaS platform.
 - Following software engineering best practices, secure coding standards, and Agile methodologies while contributing to production-ready AI solutions.

PROJECTS

- Virtual Pandit** Production Deployed
AI-Powered Spiritual Guidance Platform (Next.js, TypeScript, MongoDB, LangChain, Groq)
 - Built an AI-powered spiritual guidance platform leveraging Large Language Models to deliver contextual responses from Vedic knowledge sources.
 - Implemented RAG pipelines using LangChain and vector embeddings, improving contextual response relevance by over 40%.
 - Built semantic search across 1000+ scripture passages using vector embeddings and LangChain-powered retrieval.
 - Integrated external APIs for astrology, Panchang generation, and scripture services to provide real-time personalized experiences.
 - Designed serverless backend APIs using Next.js and deployed production workloads on AWS cloud infrastructure.
 - Optimized performance using Server-Side Rendering and Incremental Static Regeneration to improve SEO and page responsiveness.
- WinkWear** Production Deployed
AI-Enabled MERN E-commerce Platform (React, Node.js, MongoDB, Gemini, AWS)
 - Deployed and maintained a production-grade e-commerce platform on AWS, continuously serving users for over 6 months.

- Configured Google Search Console, sitemap indexing, and SEO optimization to improve search visibility and discoverability.
- Achieved search engine indexing with 150+ organic impressions and maintained a top-5 average search ranking position.
- Designed and secured 10+ REST APIs using JWT authentication and role-based authorization mechanisms.
- Integrated Gemini AI to generate personalized product recommendations with average response times below 2 seconds.
- Implemented contextual product retrieval and persistent cart management to enhance user shopping experience.

• **Mavelo**

Production Deployed

• *Cloud-Native Vehicle Rental Platform (React, Node.js, AWS, Docker, Kubernetes)*

- Developed a full-stack vehicle rental platform supporting dynamic search, booking workflows, inventory management, and real-time availability tracking.
- Containerized frontend and backend services using Docker and orchestrated deployments on Kubernetes for scalable and fault-tolerant operations.
- Leveraged AWS services including EC2, S3, CloudFront, IAM, and Route53 to build a secure and highly available cloud infrastructure.
- Implemented CI/CD pipelines using GitHub Actions, reducing release cycles from hours to minutes.
- Configured Prometheus and Grafana dashboards for real-time monitoring, alerting, resource tracking, and application observability.
- Enhanced platform security through JWT authentication, rate limiting, secure headers, and role-based access control mechanisms.

• **BlogVerse**

Production Deployed

• *Cloud-Native Publishing SaaS Platform (AWS EKS, Docker, Kubernetes, GitHub Actions)*

- Architected a highly available cloud-native publishing platform using a three-tier architecture with React, Express.js, MongoDB, and scalable service separation.
- Containerized and deployed 10+ microservices on Amazon EKS using Kubernetes, enabling scalable and highly available workloads.
- Automated build, security scanning, and deployment workflows using GitHub Actions, reducing manual deployment effort by 80%.
- Provisioned and configured AWS infrastructure including EKS, ECR, IAM, Application Load Balancers, and networking components.
- Integrated Prometheus, Grafana to monitor 20+ infrastructure metrics and centralized application logs.
- Applied enterprise-grade security practices including JWT authentication, WAF integration, and secure containerized deployments.

ACHIEVEMENTS & CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (SAA-C03) – 2026
- AWS Certified Cloud Practitioner (CLF-C02) – 2026
- Microsoft Certified: Azure Fundamentals (AZ-900) – 2025
- Recognized among the Best Submissions at TinyFish \$2M Pre-Accelerator Hackathon – 2026
- Finalist, Solvit Hackathon 2025
- Top 100 / 7000+ teams, Canara CyberSuraksha Hackathon 2025
- **LeetCode**: Solved 500+ Data Structures & Algorithms problems